

No	Answer		Marks						
1 a)	• GRAPHIC - A digital representation of non-text information such as a drawing, chart or photo • AUDIO - To increase the understanding of complex contents through audio explanation. • VIDEO - To present an information in a realistic situation or resembling a real situation • ANIMATION - To make a presentation livelier and present complex concept briefly and easily. Choose any one		State J1 Explain J1						
1 b)	<table><tr><th>Situation</th><th>Multimedia element</th></tr><tr><td>Malek inserted speech and music on his multimedia project to make the project become real.</td><td>Audio</td></tr><tr><td>Salman design a 3D interactive project for primary school to make a presentation more attractive</td><td>Animation</td></tr></table>		Situation	Multimedia element	Malek inserted speech and music on his multimedia project to make the project become real.	Audio	Salman design a 3D interactive project for primary school to make a presentation more attractive	Animation	J1 J1
Situation	Multimedia element								
Malek inserted speech and music on his multimedia project to make the project become real.	Audio								
Salman design a 3D interactive project for primary school to make a presentation more attractive	Animation								
1 c)	Picture Y: X-axis and Y-axis Picture Z: X-axis, Y-axis and Z-axis		J1 J1						
2 a)	<table><tr><th>Situation</th><th>Phase</th></tr><tr><td>a) Each object on the screen serves a purpose and communicates a visual message about the content to the user. The goal is to keep users oriented and draw them into the product</td><td>Pre-Production</td></tr><tr><td>b) The final step in the process is distributing your multimedia project</td><td>Post Production</td></tr></table>	Situation	Phase	a) Each object on the screen serves a purpose and communicates a visual message about the content to the user. The goal is to keep users oriented and draw them into the product	Pre-Production	b) The final step in the process is distributing your multimedia project	Post Production		J1 J1 J1 J1 J1 J1
Situation	Phase								
a) Each object on the screen serves a purpose and communicates a visual message about the content to the user. The goal is to keep users oriented and draw them into the product	Pre-Production								
b) The final step in the process is distributing your multimedia project	Post Production								

	c) Quality control goes on throughout the process. The storyboards are helpful for checking the sequencing Production										
	d) The prototype screens and provide the foundation for production Pre-Production										
	e) Illustrations, graphics, buttons, and icons are created using the prototype screens as a guide Production										
	f) The original files, including audio, video, and the native software formats, are archived for future upgrades or revisions Post Production										
2 b)	<table> <tr> <th>Description</th> <th>Multimedia authoring tools.</th> </tr> <tr> <td>Multimedia elements or event are present and organized with the control of speed at which it plays</td> <td>Time based Authoring tool</td> </tr> <tr> <td>These tools are best used when the bulk of your content consists of elements that can be viewed individually, letting the authoring systems link</td> <td>card / page based</td> </tr> <tr> <td>The multimedia elements organize into a structural framework, displayed as flow diagrams with branching paths.</td> <td>Icon based</td> </tr> </table>		Description	Multimedia authoring tools.	Multimedia elements or event are present and organized with the control of speed at which it plays	Time based Authoring tool	These tools are best used when the bulk of your content consists of elements that can be viewed individually, letting the authoring systems link	card / page based	The multimedia elements organize into a structural framework, displayed as flow diagrams with branching paths.	Icon based	J1 J1 J1
Description	Multimedia authoring tools.										
Multimedia elements or event are present and organized with the control of speed at which it plays	Time based Authoring tool										
These tools are best used when the bulk of your content consists of elements that can be viewed individually, letting the authoring systems link	card / page based										
The multimedia elements organize into a structural framework, displayed as flow diagrams with branching paths.	Icon based										
2 c)	Audio navigation relies on auditory cues to guide users through interfaces. It is highly beneficial for individuals with visual impairments, providing them with non-visual ways to access and interact with digital content. Examples : screen readers , voice assistants.		explain J2 example J1								
2 d)											

		Offline	Online		J1 J1
		Multimedia can be played on most personal computers and may be used for video as well as any type of multimedia content which can be delivered online	Multimedia content is being delivered via the internet in an ever-increasing list of ways,		
2 e)		Access is available anywhere at any time subject to the constraints of the viewer's Internet			J2
3 a)		<u>Reduce data redundancy</u> - With a database there is no need to repeat recording the same data <u>Improve data integrity</u> - Ensures that data is correct for all files <u>Improve data security</u> - Data can be managed by authorize user only <u>Increase data consistency</u> - Any update on data immediately available to all users <u>Improve data access</u> - Every Authorize User (DBA) can access and manage data in a database <u>Improve data sharing</u> - Data can be shared over a network, and across all application in the organization Choose any 2			J2 J2
3 b)		i) Software ii) People			J1 J1

4	<table><tr><th>Statement</th><th>Database Object</th></tr><tr><td>Allow user uses form to retrieve and maintain data in a database</td><td>Form</td></tr><tr><td>Allows user to display, and generate in hardcopy form</td><td>Report</td></tr><tr><td>Specifies which records should be retrieved by specifying criteria</td><td>Query</td></tr><tr><td>To arrange data in rows and columns and describe the characteristics of data</td><td>Table</td></tr></table>	Statement	Database Object	Allow user uses form to retrieve and maintain data in a database	Form	Allows user to display, and generate in hardcopy form	Report	Specifies which records should be retrieved by specifying criteria	Query	To arrange data in rows and columns and describe the characteristics of data	Table	J1 J1 J1 J1
Statement	Database Object											
Allow user uses form to retrieve and maintain data in a database	Form											
Allows user to display, and generate in hardcopy form	Report											
Specifies which records should be retrieved by specifying criteria	Query											
To arrange data in rows and columns and describe the characteristics of data	Table											
5 a)	<table><tr><th>Situation</th><th>The Use of Network</th></tr><tr><td>Dr. Salmi wants to be monitoring in real time the conditions of his patients in critical illness or accident.</td><td>Telemedicine</td></tr><tr><td>Fatimah watches television program content, where a central operation provides programming to many television stations providers.</td><td>Television broadcasting</td></tr><tr><td>Razi and her family travel from Johor Bahru to Mersing for vacation. He needs to monitor objects or personal movement and get from one location to another.</td><td>GPS system</td></tr></table>	Situation	The Use of Network	Dr. Salmi wants to be monitoring in real time the conditions of his patients in critical illness or accident.	Telemedicine	Fatimah watches television program content, where a central operation provides programming to many television stations providers.	Television broadcasting	Razi and her family travel from Johor Bahru to Mersing for vacation. He needs to monitor objects or personal movement and get from one location to another.	GPS system	J1 J1 J1		
Situation	The Use of Network											
Dr. Salmi wants to be monitoring in real time the conditions of his patients in critical illness or accident.	Telemedicine											
Fatimah watches television program content, where a central operation provides programming to many television stations providers.	Television broadcasting											
Razi and her family travel from Johor Bahru to Mersing for vacation. He needs to monitor objects or personal movement and get from one location to another.	GPS system											
5 b) i)	Wired Transmission Media - used a cable or other tangible material to send communication signals Wireless Transmission Media - send communications signals through the air or space / because of the absence of physical boundaries	J2 J2										

5 b) ii)	<table><tr><td>Y</td><td>Router: communications device that connects multiple computers or other routers together and transmits data to its correct destination on a network.</td></tr><tr><td>X</td><td>Wireless access point: network device used to grant network access to wireless client devices</td></tr></table>	Y	Router: communications device that connects multiple computers or other routers together and transmits data to its correct destination on a network.	X	Wireless access point: network device used to grant network access to wireless client devices	J2 J2				
Y	Router: communications device that connects multiple computers or other routers together and transmits data to its correct destination on a network.									
X	Wireless access point: network device used to grant network access to wireless client devices									
5 c)	<table><tr><th>Situation</th><th>Wired Transmission Media</th></tr><tr><td>Shahrul designed a Local Area Network for their school lab, using the cable to reduce noise</td><td>Twisted Pair Cable</td></tr><tr><td>Aminah needs to connect to her satellite television to watch his favorite movie using the ASTRO channel</td><td>Coaxial Cable</td></tr><tr><td>Dania wants to connect his home internet for a new Unifi subscriber to a better and faster internet connection</td><td>Fibre Optic Cable</td></tr></table>	Situation	Wired Transmission Media	Shahrul designed a Local Area Network for their school lab, using the cable to reduce noise	Twisted Pair Cable	Aminah needs to connect to her satellite television to watch his favorite movie using the ASTRO channel	Coaxial Cable	Dania wants to connect his home internet for a new Unifi subscriber to a better and faster internet connection	Fibre Optic Cable	J1 J1 J1
Situation	Wired Transmission Media									
Shahrul designed a Local Area Network for their school lab, using the cable to reduce noise	Twisted Pair Cable									
Aminah needs to connect to her satellite television to watch his favorite movie using the ASTRO channel	Coaxial Cable									
Dania wants to connect his home internet for a new Unifi subscriber to a better and faster internet connection	Fibre Optic Cable									
5 d)	<table><tr><th>Description</th><th>Wireless Transmission Media</th></tr><tr><td>The navigation system consists of one or more earth-based receivers that accept and analyse signals sent by satellites in order to determine the receiver's geographic location.</td><td>Microwave and Satellite transmission</td></tr><tr><td>The part of the invisible spectrum that is contiguous to the red end of the visible spectrum and that comprises electromagnetic radiation of wavelengths</td><td>Infrared transmission</td></tr></table>	Description	Wireless Transmission Media	The navigation system consists of one or more earth-based receivers that accept and analyse signals sent by satellites in order to determine the receiver's geographic location.	Microwave and Satellite transmission	The part of the invisible spectrum that is contiguous to the red end of the visible spectrum and that comprises electromagnetic radiation of wavelengths	Infrared transmission	J1 J1 J1		
Description	Wireless Transmission Media									
The navigation system consists of one or more earth-based receivers that accept and analyse signals sent by satellites in order to determine the receiver's geographic location.	Microwave and Satellite transmission									
The part of the invisible spectrum that is contiguous to the red end of the visible spectrum and that comprises electromagnetic radiation of wavelengths	Infrared transmission									

	<table><tr><td>A mobile phone uses high-frequency radio waves to transmit voice and digital data messages.</td><td>Cellular radio transmission</td></tr></table>	A mobile phone uses high-frequency radio waves to transmit voice and digital data messages.	Cellular radio transmission							
A mobile phone uses high-frequency radio waves to transmit voice and digital data messages.	Cellular radio transmission									
5 e)	Network adapter/ Network interface card	J1								
6 a)	<table><tr><td>Description</td><td>Type of search strategies</td></tr><tr><td>To further specify exactly what you want a search engine to find by using AND, OR and NOT.</td><td>Boolean operators</td></tr><tr><td>Limits the search to a particular search characteristic, such as the page title, URL, page text, top-level domain, or Web site.</td><td>field searches</td></tr><tr><td>Most search engines automatically list the hits that include all the keywords first, followed by hits matching most of the keywords, continuing down to hits that fit only one of the keywords.</td><td>Phrases strategies</td></tr></table>	Description	Type of search strategies	To further specify exactly what you want a search engine to find by using AND, OR and NOT.	Boolean operators	Limits the search to a particular search characteristic, such as the page title, URL, page text, top-level domain, or Web site.	field searches	Most search engines automatically list the hits that include all the keywords first, followed by hits matching most of the keywords, continuing down to hits that fit only one of the keywords.	Phrases strategies	J1 J1 J1
Description	Type of search strategies									
To further specify exactly what you want a search engine to find by using AND, OR and NOT.	Boolean operators									
Limits the search to a particular search characteristic, such as the page title, URL, page text, top-level domain, or Web site.	field searches									
Most search engines automatically list the hits that include all the keywords first, followed by hits matching most of the keywords, continuing down to hits that fit only one of the keywords.	Phrases strategies									

6 b)	<table border="1"> <thead> <tr> <th>Scenario</th><th>web activities</th><th>Example</th></tr> </thead> <tbody> <tr> <td>i) Dayang used the web for listening the songs, watching videos and movies</td><td>Online Entertainment</td><td>Netflix / iTune / Joox, Spotify</td></tr> <tr> <td>ii) Hizman works as a supervisor at Sendiri Maju Company, he sends reports about the performance of his staff to the Manager in Kuala Lumpur.</td><td>Online communications</td><td>e-mail</td></tr> <tr> <td>iii) Noryn likes to share their personal or career content, interests, activities, backgrounds or real-life connections.</td><td>Social Networking/social media</td><td>Instagram / Facebook (other Online social media)</td></tr> <tr> <td>iv) Nor Aminah is an influence, she wants her followers buy the products on online services or over the Internet</td><td>E- commerce / online shopping</td><td>shopee.com, Lazada.com mudah.com</td></tr> </tbody> </table>	Scenario	web activities	Example	i) Dayang used the web for listening the songs, watching videos and movies	Online Entertainment	Netflix / iTune / Joox, Spotify	ii) Hizman works as a supervisor at Sendiri Maju Company, he sends reports about the performance of his staff to the Manager in Kuala Lumpur.	Online communications	e-mail	iii) Noryn likes to share their personal or career content, interests, activities, backgrounds or real-life connections.	Social Networking/social media	Instagram / Facebook (other Online social media)	iv) Nor Aminah is an influence, she wants her followers buy the products on online services or over the Internet	E- commerce / online shopping	shopee.com, Lazada.com mudah.com	J1 J1 J1 J1
Scenario	web activities	Example															
i) Dayang used the web for listening the songs, watching videos and movies	Online Entertainment	Netflix / iTune / Joox, Spotify															
ii) Hizman works as a supervisor at Sendiri Maju Company, he sends reports about the performance of his staff to the Manager in Kuala Lumpur.	Online communications	e-mail															
iii) Noryn likes to share their personal or career content, interests, activities, backgrounds or real-life connections.	Social Networking/social media	Instagram / Facebook (other Online social media)															
iv) Nor Aminah is an influence, she wants her followers buy the products on online services or over the Internet	E- commerce / online shopping	shopee.com, Lazada.com mudah.com															
6 c) i)	search engine	J1															
6 c) ii)	•A software system designed to carry out web searches by allowing the user to enter keywords at are run against a database or •A search engine is a program that finds Web sites, Web pages, images, videos, news, maps, and other information related to a specific topic.	J2															
6 d)	Treat email as a formal communication method Don't use personal email for business Encrypt sensitive email Always enable two step verification Secure confidential attachments Choose any 2	J1 J1															

6 e)	Home automation – - alarm system to prevent thief - smart thermostat - adjust your home's temperature using a mobile app - sprinkling system to water the plant Fitness System – track daily activities Choose any 1 or any acceptable answer	J2
6 f)	Use computers and devices that comply with the ENERGY STAR program. Do not leave the computer running overnight. Turn off the monitor, printer and other devices when not in use. Recycle paper	J1 J1